

PHY 211

Lecture 11 Notes

By

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Chapter 10

Image Formation (Mirrors & Lenses)

➤ General definitions:

- Principal axis is the normal line to the mid-point of surface (Vertex V or pole P)
- Object location (O) is the position of the object on the principal axis.
- The object distance (p) is the distance from the object to the mirror or lens.
- Image location (I) is the position of the image on the principal axis.
- The image distance (q) is the distance from the image to the mirror or lens.
- Object size (h) is the height of the object.
- Image size (h') is the height of the image (+ve for upright image “same direction as object” & -ve for inverted image “opposite direction of object”).
- Magnification (M) is the ratio between image and object size:

$$M = \frac{h'}{h}$$

- Focal point “Focus” (F) is the point where parallel rays reflect or refract to pass through or appear to originate from.
- Focal length (f) is the distance between focal point and mirror or lens.
- Center of curvature (C) is the center of the sphere from which the surface is cut from.
- Radius of curvature (R) is the radius of the sphere from which the surface is cut from (distance between C & surface)
- For mirrors: $R = 2f$
- For far away sources (infinity), the rays are considered to be parallel (plane waves)

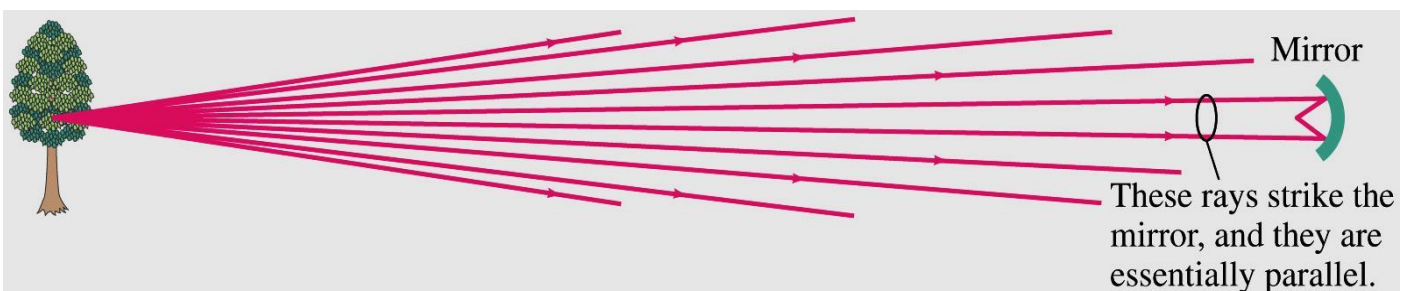


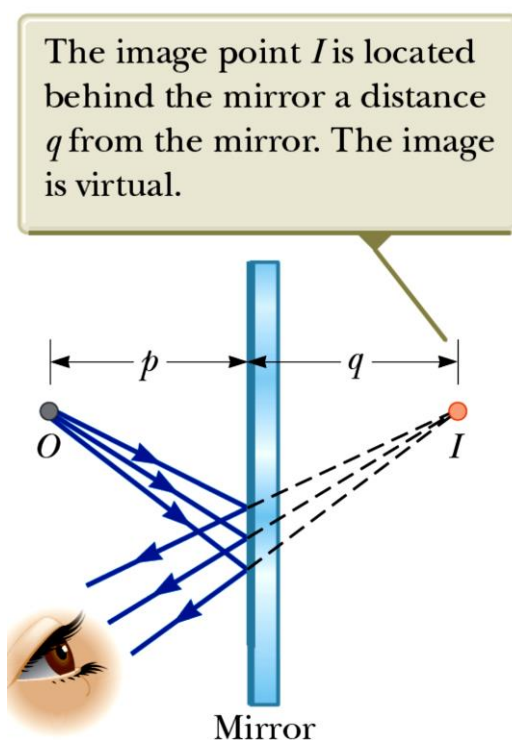
Image	
Real image	Virtual image
Formed from the intersection of the reflected rays for mirrors and refracted rays for lenses.	Formed from the intersection of the <u>extensions</u> of reflected rays for mirrors and refracted rays for lenses
Can be viewed on screen.	Cannot be viewed on screen
Formed in front of mirrors and behind “back of” lenses.	Formed behind (back of) mirrors and in front of lenses.
Image distance (q) is +ve	Image distance (q) is -ve

Mirrors

1- Plane mirrors

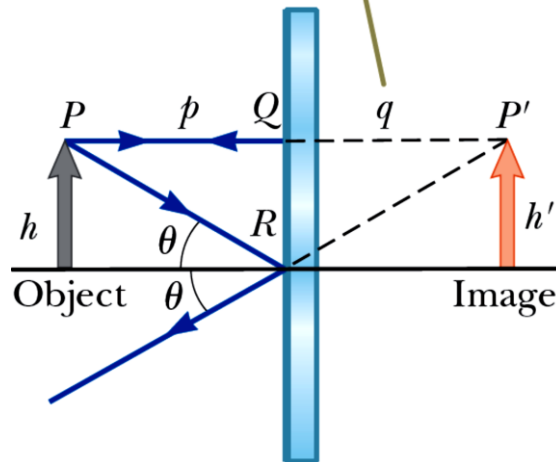
2- Spherical mirrors (concave and convex mirrors)

1. Plane mirrors:



- Consider a point source of light placed at O . Light rays leave the source and are reflected from the mirror.
- After reflection, the rays diverge (spread apart), but they appear to the viewer to come from a point I behind the mirror. Point I is called the image of the object at O .
- The image formed by the flat mirror is a virtual image (q is $-ve$).
- In fact, the images seen in flat mirrors are always virtual (for real objects).

Because the triangles PQR and $P'QR$ are congruent,
 $|p| = |q|$ and $h = h'$.



- One of those rays starts at P , follows the horizontal path PQ to the mirror, and reflects back on itself. The second ray follows the path PR and reflects as shown obeying reflection law.
- An observer to the left of the mirror would trace the two reflected rays back to the point from which they appear to have originated: point P' .
- A continuation of this process for points other than P on the object would result in a virtual image to the right of the mirror.
- Because triangles PQR and $P'QR$ are identical, $PQ = P'Q$.

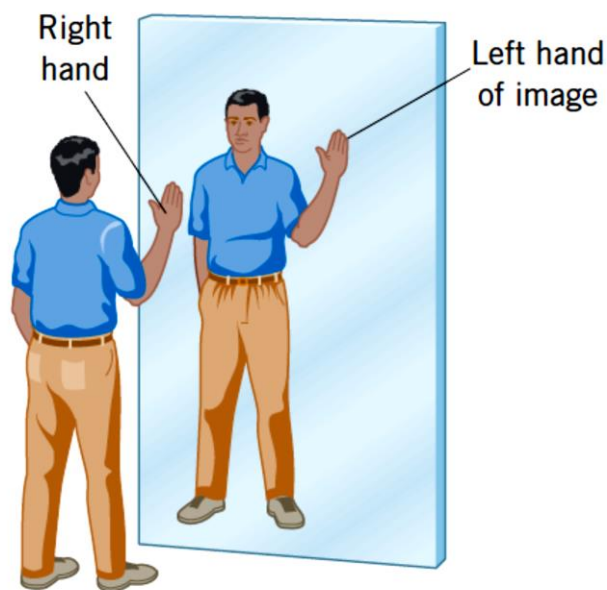
$$|p| = |q| \text{ or } p = -q$$

$$M = -\frac{q}{p} = \frac{h'}{h} = 1 \text{ (For any flat mirror)}$$

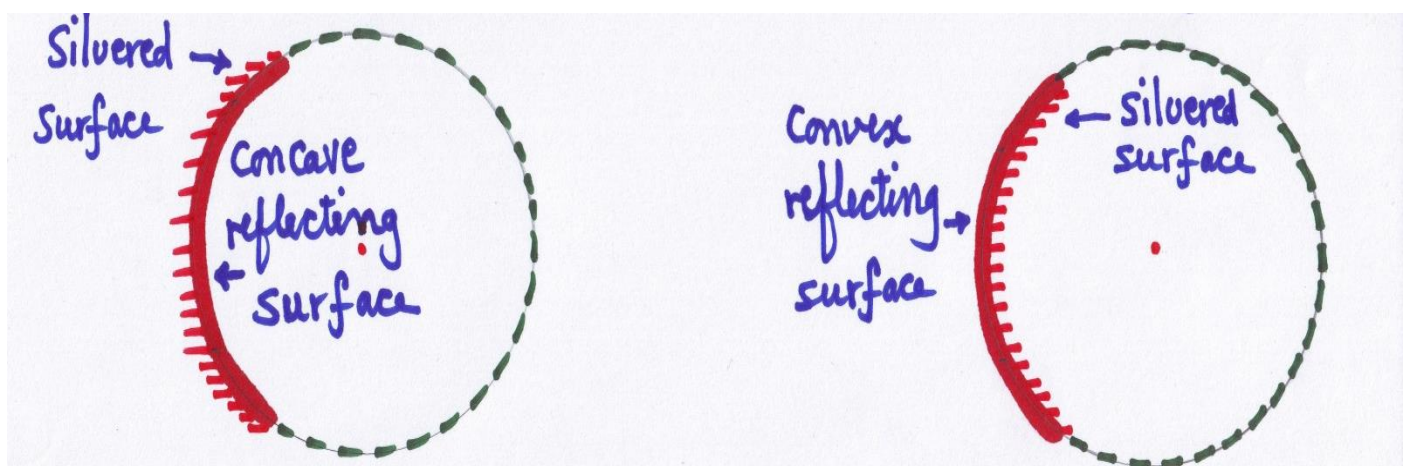
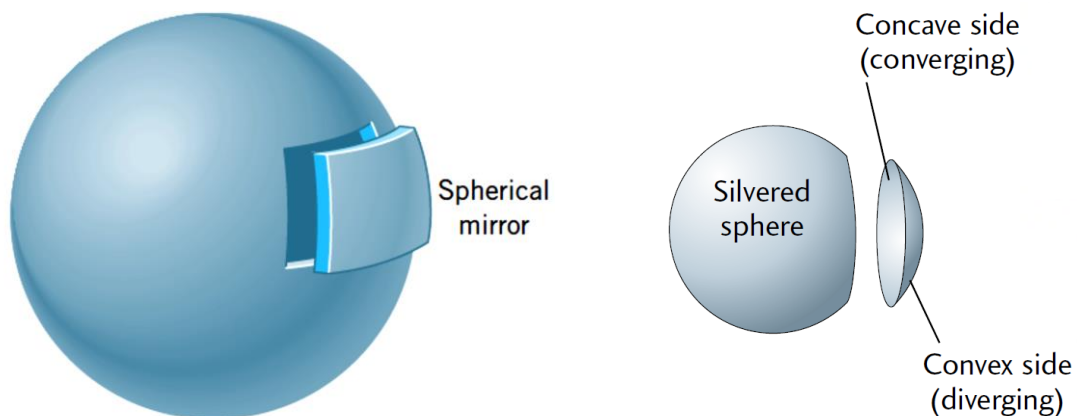
- The image formed by a flat mirror has the following properties:
 1. **The image is as far behind the mirror as the object is in front** ($|p| = |q|$).
 2. **The image is unmagnified** ($M = 1$ & $|h'| = |h|$), **virtual** (q is $-ve$), and **upright** (M & h' are $+ve$).

(By *upright*, we mean that if the object arrow points upward, as in figure, so does the image arrow. The opposite of an upright image is an inverted image.)

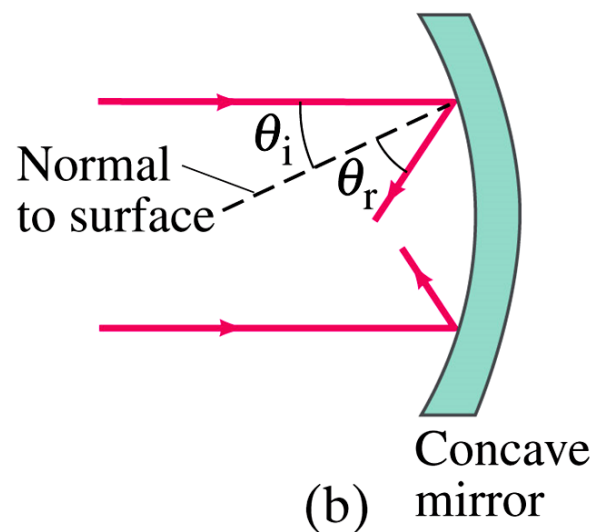
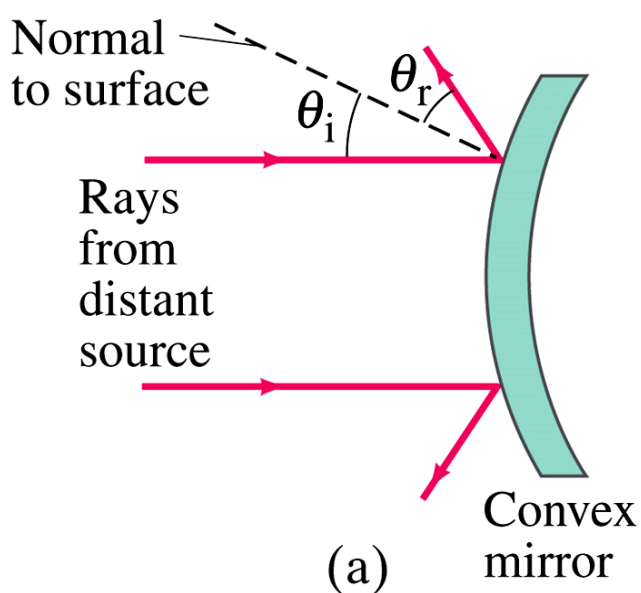
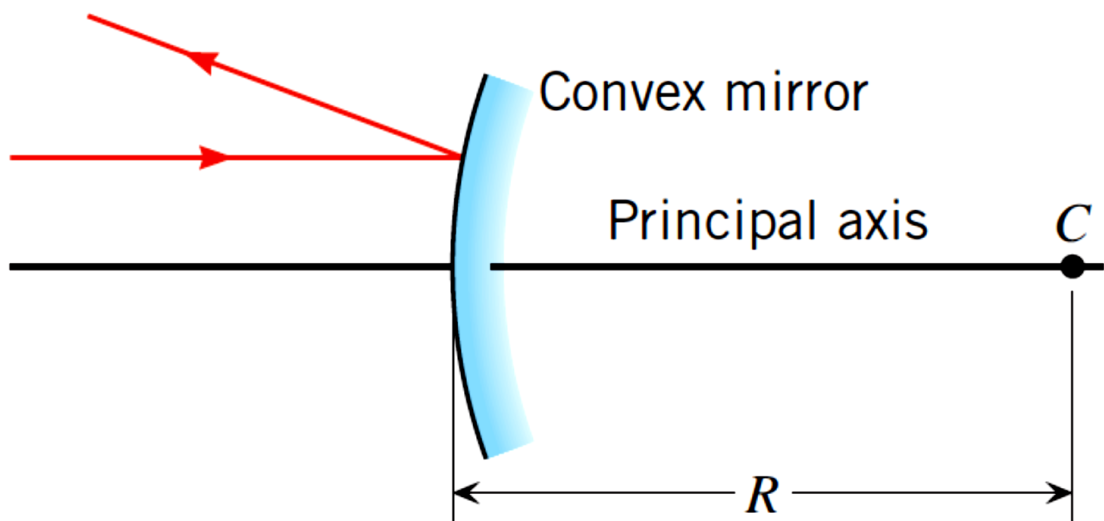
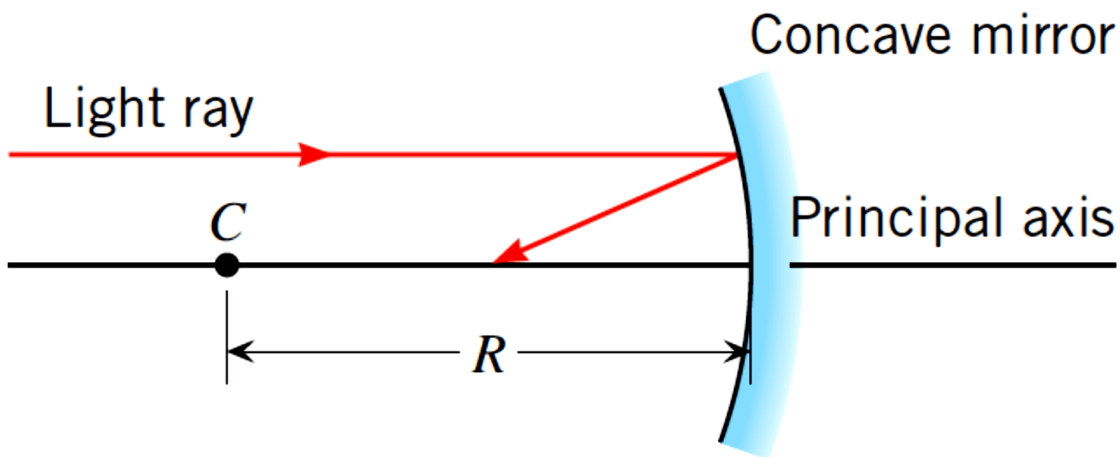
- Note that a flat mirror produces an image having an **apparent left–right reversal**. You can see this reversal standing in front of a mirror and raising your right hand.

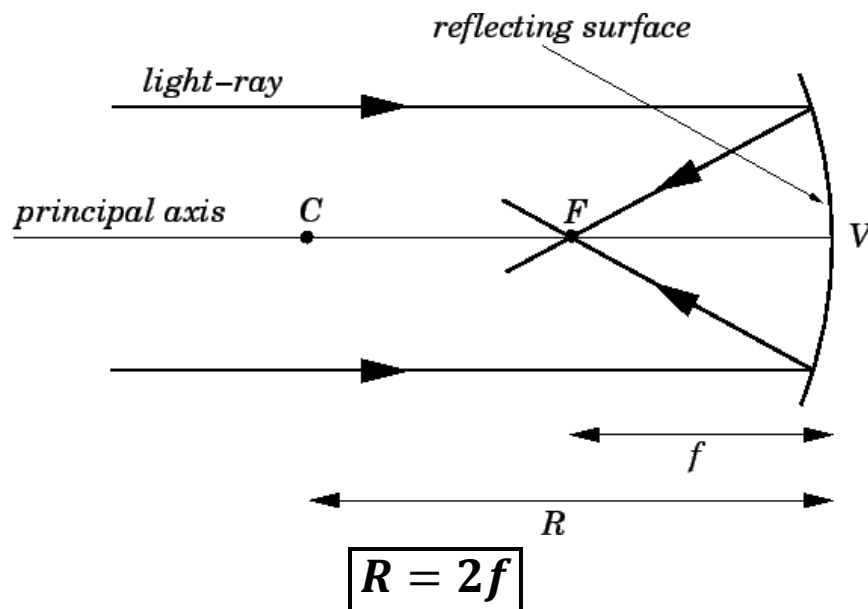


2. Spherical mirrors

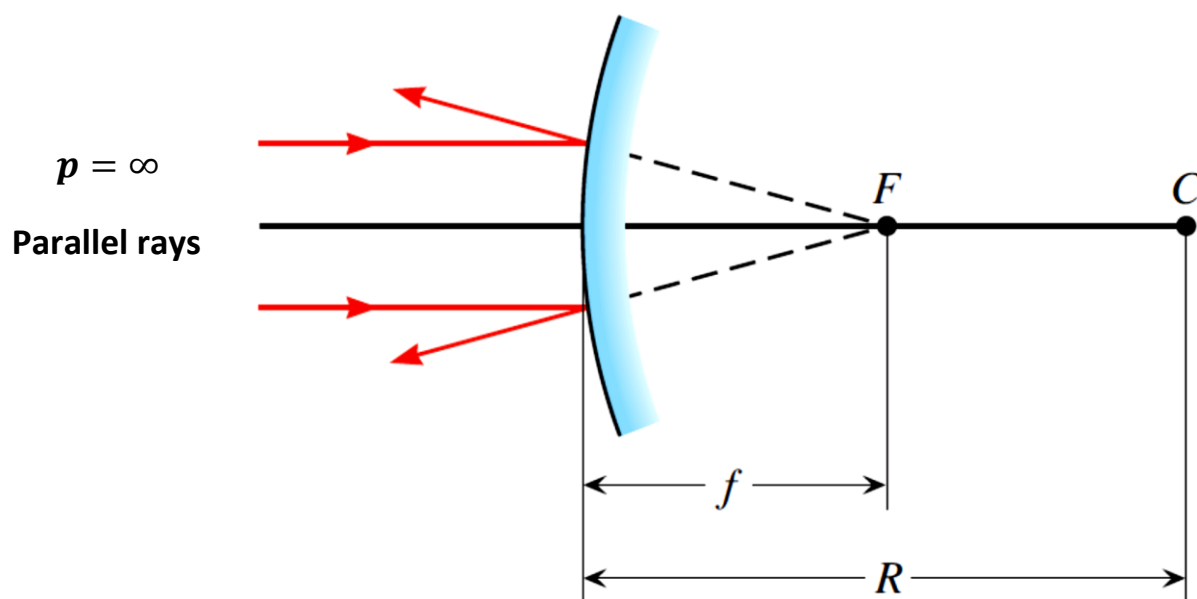
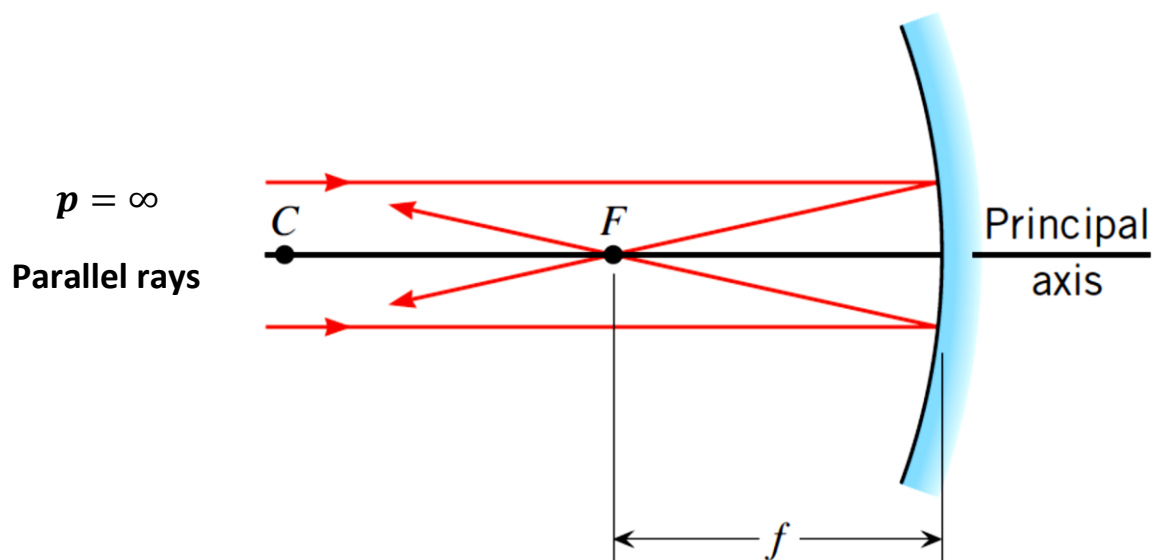


- Concave mirror is converging mirror & convex mirror is diverging mirror



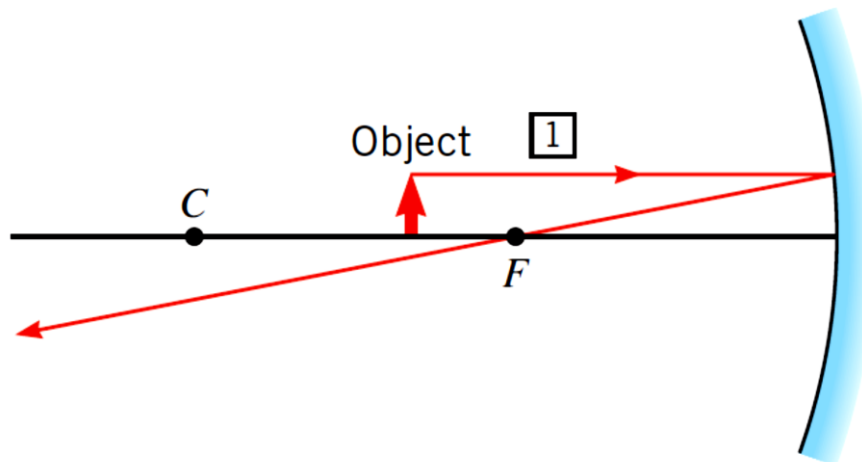


- Focal point & center of curvature are in front of concave mirror & in the back of convex mirror

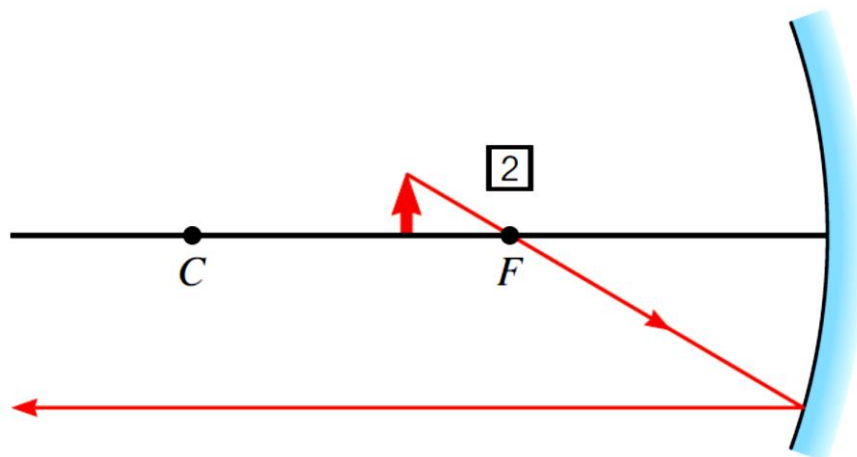


- **Ray Diagrams for Mirrors**

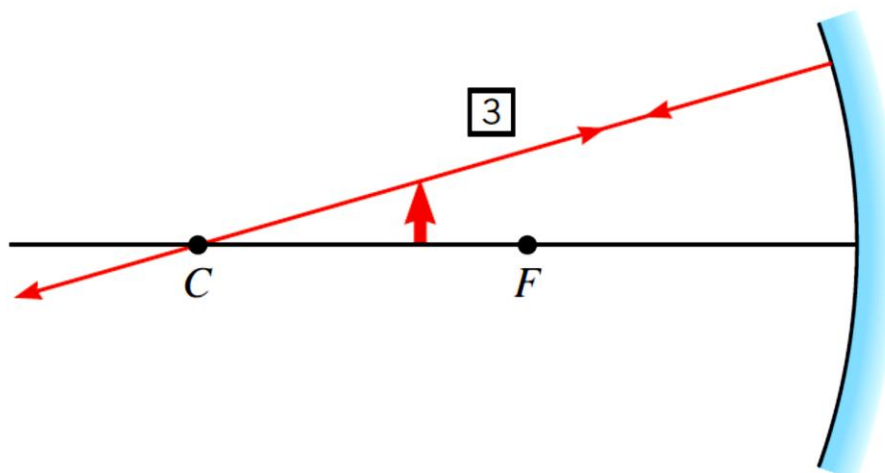
1. Ray 1 is drawn parallel to the principal axis and is reflected back through the focal point F .



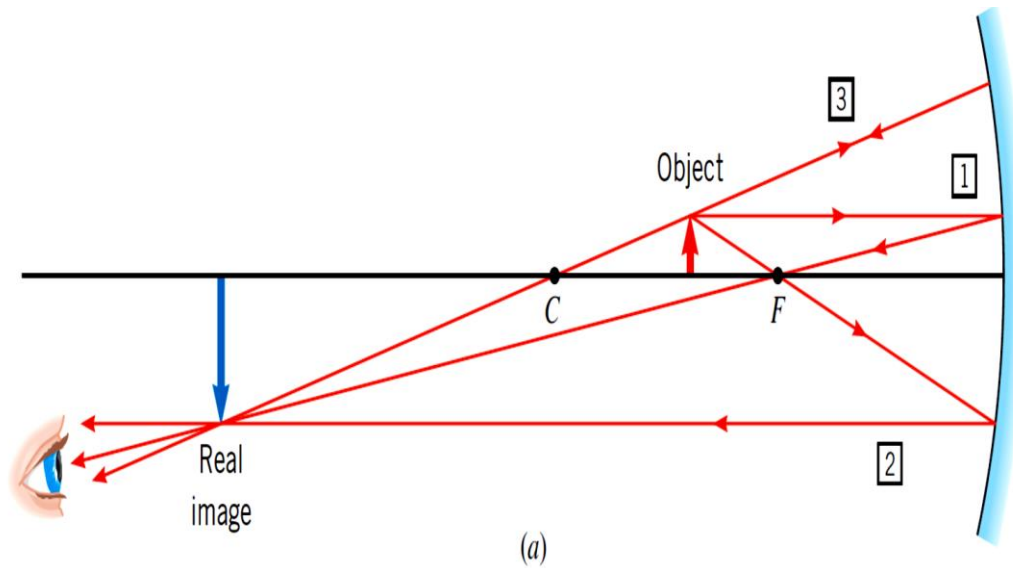
2. Ray 2 is drawn through the focal point and is reflected parallel to the principal axis.



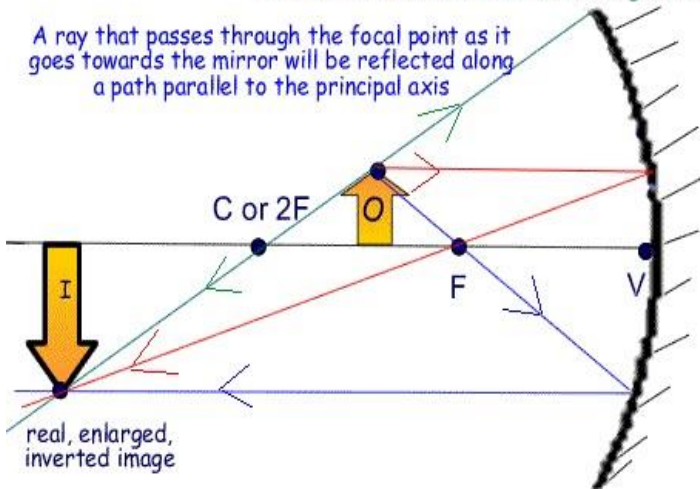
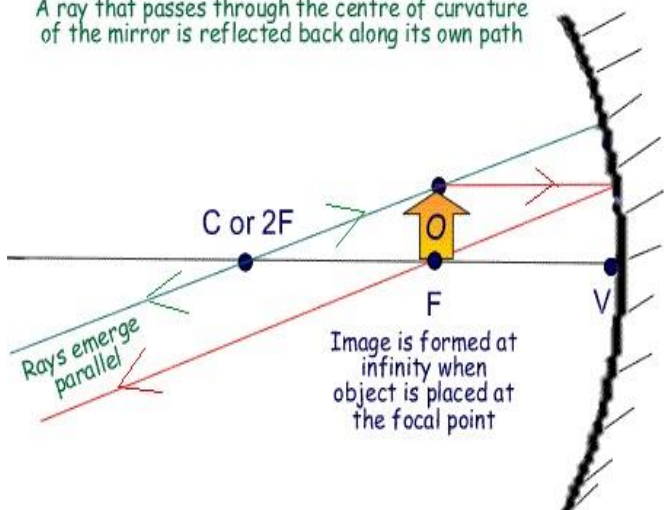
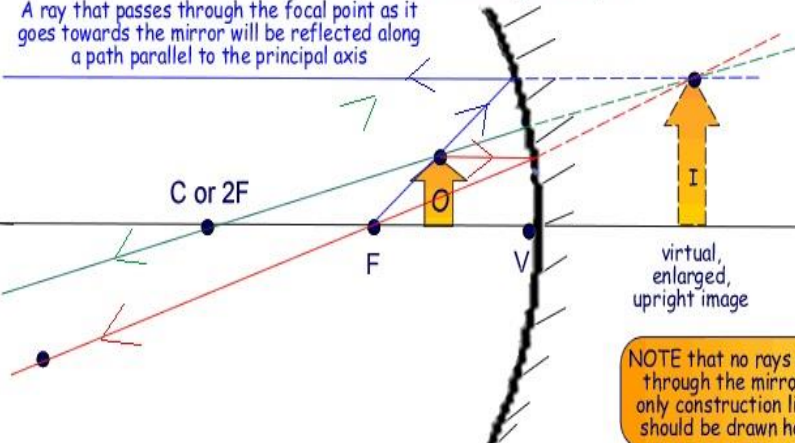
3. Ray 3 is drawn through the center of curvature, C , and is reflected back on itself.



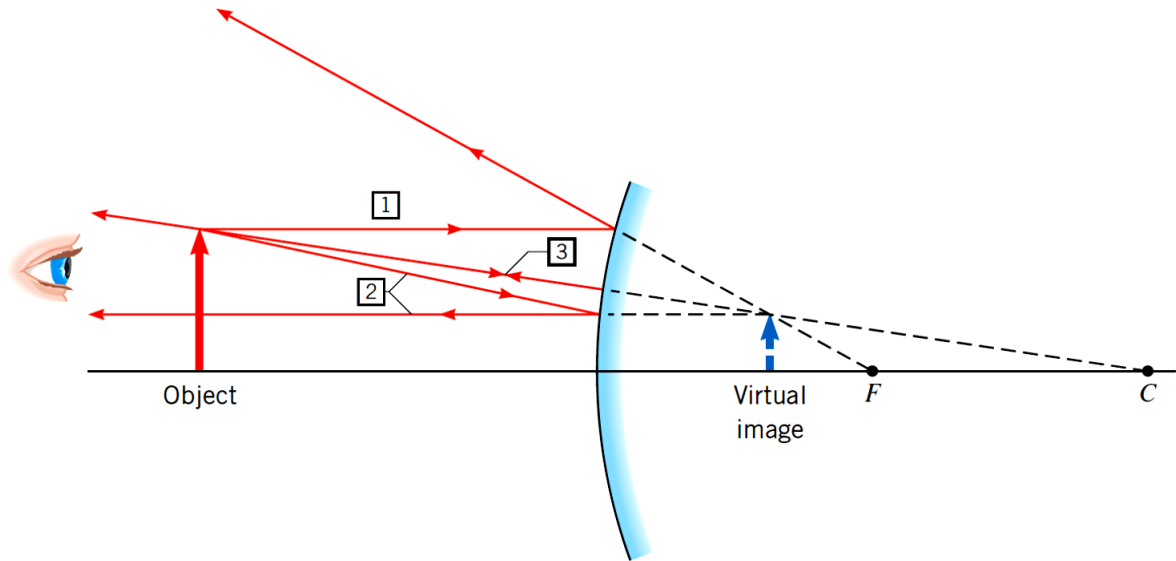
- For concave mirror:



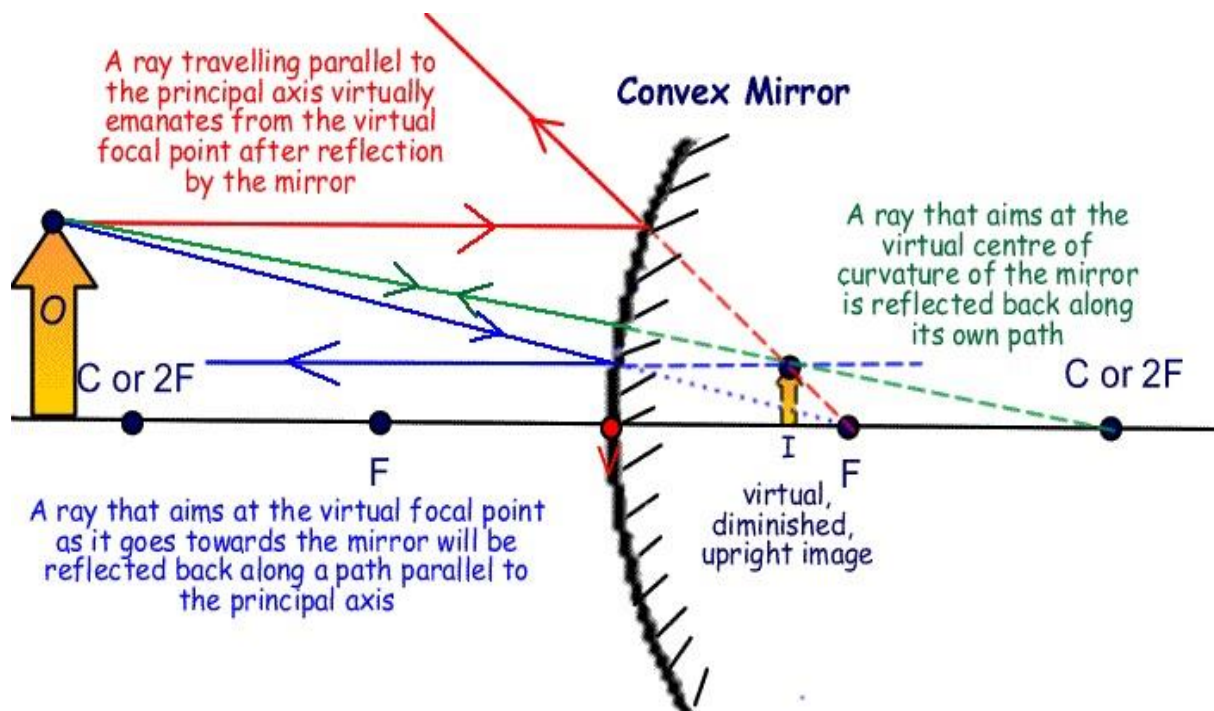
Object distance	Image characteristics	Ray diagram
$p > 2f$	Real, inverted & reduced	A ray travelling parallel to the principal axis passes through the focal point after reflection by the mirror A ray that passes through the centre of curvature of the mirror is reflected back along its own path A ray that passes through the focal point as it goes towards the mirror will be reflected along a path parallel to the principal axis real, diminished, inverted image
$p = 2f$	Real, inverted & same size ($q = p = 2f$)	A ray travelling parallel to the principal axis passes through the focal point after reflection by the mirror A ray that passes through the focal point as it goes towards the mirror will be reflected along a path parallel to the principal axis real, inverted image - the same size as the object

$2f > p > f$	Real, inverted & enlarged	A ray travelling parallel to the principal axis passes through the focal point after reflection by the mirror A ray that passes through the centre of curvature of the mirror is reflected back along its own path A ray that passes through the focal point as it goes towards the mirror will be reflected along a path parallel to the principal axis  real, enlarged, inverted image
$p = f$	Image formed at infinity	A ray travelling parallel to the principal axis passes through the focal point after reflection by the mirror A ray that passes through the centre of curvature of the mirror is reflected back along its own path  Rays emerge parallel Image is formed at infinity when object is placed at the focal point
$p < f$	Virtual, upright and enlarged	A ray travelling parallel to the principal axis passes through the focal point after reflection by the mirror A ray that passes through the centre of curvature of the mirror is reflected back along its own path A ray that passes through the focal point as it goes towards the mirror will be reflected along a path parallel to the principal axis  virtual, enlarged, upright image NOTE that no rays pass through the mirror - only construction lines should be drawn here

- For convex mirror



Object distance	Image characteristics
Anywhere in front of mirror	Virtual, upright & reduced



- Mirror equation:

$$\boxed{\frac{1}{p} + \frac{1}{q} = \frac{2}{R}} \quad \text{mirror equation}$$

$$\therefore R = 2f \longrightarrow \boxed{\frac{1}{p} + \frac{1}{q} = \frac{1}{f}}$$

The mirror power is given by:

$$\boxed{F(\Delta \text{ or } m^{-1}) = \frac{1}{f(m)} = \frac{2}{R(m)}} \quad \Delta: \text{Diopter}$$

- Magnification:

$$\boxed{M = -\frac{q}{p}}$$

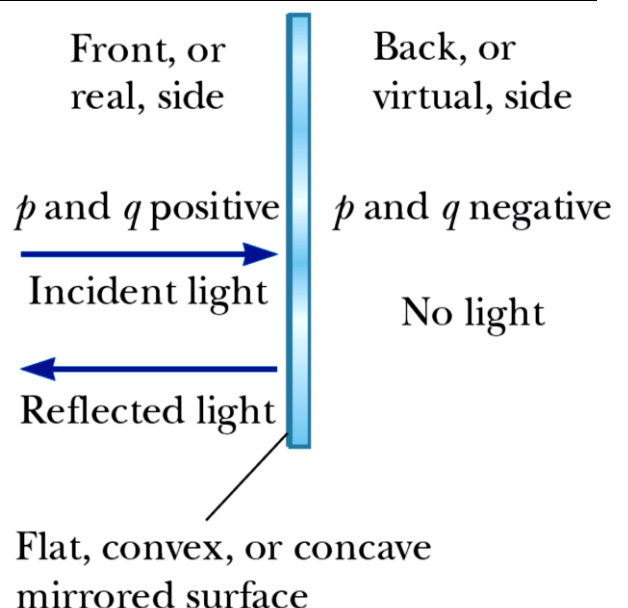
- Sign convection for the mirrors:

	Positive	Negative
Object location (p)	Real Object (in front of mirror)	Virtual object (behind mirror)
Image location (q)	Real image (in front of mirror)	Virtual image (behind mirror)
Image height (h')	Image is upright	Image is inverted
Magnification (M)	Image is upright	Image is inverted
Focal length (f) and radius of curvature (R)	Concave mirror	Convex mirror

$|M| = 1$ *image size = object size*

$|M| > 1$ *image is enlarged*

$|M| < 1$ *image is diminished*



- **Example 1:** Assume a certain concave, spherical mirror has a focal length of +10.0 cm. Locate and describe the image for object distances
 - 25 cm
 - 10.0 cm and
 - 5.00 cm.

Solution

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

a)

$$\frac{1}{25} + \frac{1}{q} = \frac{1}{10} \longrightarrow \frac{1}{q} = \frac{1}{10} - \frac{1}{25} = \frac{3}{50}$$

$$q = \frac{50}{3} = 16.7 \text{ cm } (+ve: \text{real image})$$

$$M = \frac{-q}{p} = \frac{-\frac{50}{3}}{25} = -\frac{2}{3} = -0.668 \text{ } (-ve: \text{inverted}, |M| < 1: \text{diminished})$$

The image is real, inverted and diminished

b)

$$\frac{1}{10} + \frac{1}{q} = \frac{1}{10} \longrightarrow \frac{1}{q} = \frac{1}{10} - \frac{1}{10} = 0$$

$$q = \infty \text{ (image at infinity)}$$

c)

$$\frac{1}{5} + \frac{1}{q} = \frac{1}{10} \longrightarrow \frac{1}{q} = \frac{1}{10} - \frac{1}{5} = \frac{-1}{10}$$

$$q = -10. \text{ cm } (-ve: \text{virtual image})$$

$$M = \frac{-q}{p} = \frac{-(-10)}{5} = \frac{10}{5} = 2 (+ve: \text{upright}, |M| > 1: \text{enlarged})$$

The image is virtual, upright and enlarged